YEAR 10 MATHS · CHAPTER 1 · MIND MAP

Algebra, equations and linear relationships

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1 Review of algebra

Collect like terms; expand by multiplying every term; factorise by taking out the HCF.

$5a - 2b + 3a = 8a - 2b$

$3x(2 - x) = 6x - 3x^2$

$6ab - 9a = 3a(2b - 3)$

2 Solving linear equations

Undo each operation in reverse order, doing the same to both sides.

$\frac{2x+1}{3} = 5 \Rightarrow 2x + 1 = 15 \Rightarrow x = 7$

$5(x - 1) = 2x + 10 \Rightarrow 3x = 15 \Rightarrow x = 5$

3 Linear inequalities

Solve like an equation, but flip the sign when you multiply or divide by a negative.

$3 - 2x \leq 11 \Rightarrow -2x \leq 8 \Rightarrow x \geq -4$

Watch out: flip the sign, not just the number.

4 Equations with algebraic fractions

Multiply every term by the lowest common denominator first.

$\frac{x}{3} + \frac{x-1}{2} = 4$: $\times 6$, $2x + 3(x - 1) = 24$, $x = 5.4$

$\frac{x+2}{4} = \frac{x-1}{3} \Rightarrow 3x + 6 = 4x - 4 \Rightarrow x = 10$

5 Graphing straight lines

$y = mx + c$. Or find the two intercepts and join them.

$3x - 2y = 6$: x-intercept $(2, 0)$, y-intercept $(0, -3)$

$y = -\frac{1}{2}x + 4$: gradient $-\frac{1}{2}$, y-intercept 4

6 Finding the equation of a line

Find the gradient, then use $y - y_1 = m(x - x_1)$.

Through $(2, 3)$ and $(4, 11)$: $m = \frac{8}{2} = 4$

$y - 3 = 4(x - 2) \Rightarrow y = 4x - 5$

7 Length and midpoint

Midpoint: average the coordinates. Length: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$

$(-1, 2)$ and $(5, 10)$: midpoint $(2, 6)$

Length $\sqrt{6^2 + 8^2} = 10$

8 Parallel and perpendicular lines

Parallel: equal gradients. Perpendicular: $m_1 \times m_2 = -1$.

Perpendicular to $y = 3x + 1$: gradient $-\frac{1}{3}$

Through $(3, 2)$: $y = -\frac{1}{3}x + 3$

9 Simultaneous equations: substitution

Replace one pronumeral with what it equals from the other equation.

$y = 2x - 1$ and $3x + y = 14$

$3x + 2x - 1 = 14 \Rightarrow x = 3,\ y = 5$

10 Simultaneous equations: elimination

Add or subtract the equations so one pronumeral cancels.

$2x + 3y = 12$ and $4x - 3y = 6$

Add: $6x = 18$, so $x = 3,\ y = 2$

11 Applications of simultaneous equations

Define two pronumerals, write two equations, solve, then answer in words.

Two numbers add to 31 and differ by 7: $x + y = 31,\ x - y = 7$

Add: 2x = 38, so the numbers are **19** and **12**

12 Regions on the Cartesian plane

Draw the boundary line (dashed for $<$ or $>$, solid for $\leq$ or $\geq$), then test a point.

$y > 2x + 1$: test $(0, 0)$: $0 > 1$ is false

So shade the side without the origin

YEAR 10 MATHS · CHAPTER 2 · MIND MAP

Geometry and networks

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1 Review of geometry

Interior angle sum of an n-gon: $(n-2) \times 180°$. Exterior angles add to 360°.

Regular octagon: exterior $360° \div 8 = 45°$, interior $135°$

2 Congruent triangles and quadrilaterals

Tests: SSS, SAS, AAS, RHS. Use them to prove properties of shapes.

In a parallelogram, the diagonals cut it into congruent triangles (AAS), so the diagonals bisect each other

Watch out: AAA shows similar, not congruent.

3 Similar figures and triangles

Scale factor k: lengths $\times k$, areas $\times k^2$, volumes $\times k^3$.

$k = 3$: area $\times 9$, volume $\times 27$

Similar triangles, matching sides 4 and 10: $k = 2.5$, so side 6 matches 15

4 Circles and chords

A line from the centre at right angles to a chord bisects the chord.

Chord 16, radius 10: distance from the centre $\sqrt{10^2 - 8^2} = 6$

Chord, arc, sector, segment, tangent, secant: know each word.

5 Angle properties of circles

Angle at the centre is twice the angle at the circumference. Angles in the same segment are equal.

Angle in a semicircle = 90°

Cyclic quadrilateral: opposite angles add to 180° (110° opposite 70°)

Circumference angle 35°: centre angle 70°

6 Circles and tangents

A tangent meets the radius at 90°. Two tangents from one point are equal.

Radius 5, centre 13 away: tangent $\sqrt{13^2 - 5^2} = 12$

Alternate segment theorem: tangent–chord angle equals the angle in the opposite segment.

7 Chords, secants and tangents

Chords: $AP \times PB = CP \times PD$. Tangent: $PT^2 = PA \times PB$.

$4 \times 6 = 3 \times x \Rightarrow x = 8$

$PA = 4,\ PB = 9$: $PT = \sqrt{36} = 6$

8 Introduction to networks

Vertices (dots) joined by edges. The degree of a vertex is the number of edges at it.

The degrees always add to twice the number of edges

5 edges: degrees add to **10**

9 Isomorphic and planar graphs

Isomorphic: same connections, drawn differently. Planar: can be drawn with no crossing edges.

Euler's formula for planar graphs: $v - e + f = 2$

4 vertices, 6 edges: $f = 2 - 4 + 6 = 4$ faces

10 Trails, paths and Euler circuits

Trail: no repeated edges. Path: no repeated vertices.

Euler circuit: every vertex has an even degree

Euler trail: exactly two odd vertices (start at one, finish at the other)

11 Shortest path problems

Add the weights along each route and keep the smallest at every vertex.

A to B 4, B to D 3, A to C 2, C to D 6

A–B–D = 7, A–C–D = 8, so the shortest is **7**

YEAR 10 MATHS · CHAPTER 3 · MIND MAP

Indices, exponentials and logarithms

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1 Review of index laws

$a^m a^n = a^{m+n}$, $a^m \div a^n = a^{m-n}$, $(a^m)^n = a^{mn}$, $a^0 = 1$

$(2x^3)^2 \times x = 4x^6 \times x = 4x^7$

2 Negative indices

$a^{-n} = \frac{1}{a^n}$

$3^{-2} = \frac{1}{9}$ $2x^{-3} = \frac{2}{x^3}$

$\left(\frac{2}{3}\right)^{-2} = \frac{9}{4}$

Watch out: only the x is to the power -3 in $2x^{-3}$.

3 Scientific notation

$a \times 10^n$ with $1 \le a < 10$.

$0.000\,52 = 5.2 \times 10^{-4}$

$(3 \times 10^5)(4 \times 10^{-2}) = 12 \times 10^3 = 1.2 \times 10^4$

4 Fractional indices

$a^{\frac{1}{n}} = \sqrt[n]{a}$ and $a^{\frac{m}{n}} = (\sqrt[n]{a})^m$

$27^{\frac{2}{3}} = 3^2 = 9$ $16^{-\frac{1}{2}} = \frac{1}{4}$

5 Exponential equations

Write both sides with the same base, then equate the indices.

$2^x = 32 = 2^5 \Rightarrow x = 5$

$9^x = 27 \Rightarrow 3^{2x} = 3^3 \Rightarrow x = \frac{3}{2}$

6 Exponential graphs

$y = a^x$ $(a > 1)$ passes through $(0, 1)$ and gets close to $y = 0$ but never reaches it.

$y = 2^x$: $(0, 1), (1, 2), (2, 4)$

$y = 2^{-x}$ is its reflection in the y-axis

7 Exponential growth and decay

Growth: $A = A_0(1 + r)^n$. Decay: $A = A_0(1 - r)^n$.

2000 people growing 5% a year: $2000 \times 1.05^3 \approx 2315$ after 3 years

Decay 10% a year: multiply by 0.9 each year

8 Compound interest

$A = P\left(1 + \frac{r}{k}\right)^{nk}$: k payments a year for n years.

$5000 at 6% p.a. compounded monthly for 2 years: $5000 \times 1.005^{24} \approx \$5635.80$

Watch out: divide the yearly rate by 12 for monthly.

9 Introducing logarithms

$\log_a b = x$ means $a^x = b$.

$\log_2 8 = 3$ $\log_{10} 1000 = 3$ $\log_3 \frac{1}{9} = -2$

10 Logarithmic scales

Each step up the scale is 10 times bigger.

A magnitude 6 earthquake has 10 times the amplitude of a magnitude 5

pH 3 is 10 times more acidic than pH 4

11 Laws of logarithms

$\log xy = \log x + \log y$, $\log \frac{x}{y} = \log x - \log y$, $\log x^n = n \log x$

$\log_2 12 - \log_2 3 = \log_2 4 = 2$

$\log_a 1 = 0$ $\log_a a = 1$

12 Solving with logarithms

Take logs of both sides when you can't match the bases.

$3^x = 20 \Rightarrow x = \frac{\log 20}{\log 3} \approx 2.73$

Check: $3^{2.73} \approx 20$ ✓

YEAR 10 MATHS · CHAPTER 4 · MIND MAP

Measurement and surds

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1 Irrational numbers and surds

A surd is a root that isn't a whole number, like $\sqrt{2}$. Simplify using square factors.

$\sqrt{8} = 2\sqrt{2}$ $\sqrt{48} = \sqrt{16}\sqrt{3} = 4\sqrt{3}$

2 Operations with surds

Add and subtract like surds only. Multiply or divide the numbers outside and inside separately.

$\sqrt{12} + \sqrt{27} = 2\sqrt{3} + 3\sqrt{3} = 5\sqrt{3}$

$\sqrt{6} \times \sqrt{3} = \sqrt{18} = 3\sqrt{2}$ $6\sqrt{10} \div 2\sqrt{5} = 3\sqrt{2}$

3 Rationalising the denominator

Multiply top and bottom so there's no surd on the bottom.

$$\frac{3}{\sqrt{2}} = \frac{3\sqrt{2}}{2}$$

$$\frac{1}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{\sqrt{3}-1}{2}$$

4 Review of length

Circumference $2\pi r$. Arc length $\frac{\theta}{360} \times 2\pi r$.

$r = 10,\ \theta = 72°$: arc $\frac{1}{5} \times 20\pi = 4\pi \approx 12.57$

5 Pythagoras in 3D

Find a right triangle, or use $d = \sqrt{a^2 + b^2 + c^2}$ for a box.

Box 2 by 3 by 6: $\sqrt{4 + 9 + 36} = 7$

6 Review of area

Triangle $\frac{1}{2}bh$, trapezium $\frac{1}{2}(a+b)h$, circle πr^2, sector $\frac{\theta}{360}\pi r^2$

Sector $r = 6,\ \theta = 60°$: $\frac{1}{6} \times 36\pi = 6\pi \approx 18.85$

7 Measurement errors and accuracy

The error is up to half the smallest unit. Percentage error = error ÷ measurement × 100%.

4.2 m to the nearest 0.1 m: between 4.15 m and 4.25 m

Percentage error $\frac{0.05}{4.2} \times 100\% \approx 1.2\%$

8 Surface area: prisms and cylinders

Add every face. Cylinder: $2\pi r^2 + 2\pi rh$.

Cylinder $r = 4,\ h = 10$: $32\pi + 80\pi = 112\pi \approx 351.9$

9 Surface area: pyramids and cones

Cone: $\pi r^2 + \pi rs$, where s is the slant height.

$r = 3,\ h = 4$: $s = 5$, so $SA = 9\pi + 15\pi = 24\pi \approx 75.4$

Watch out: use the slant height, not the vertical height.

10 Volume: prisms and cylinders

$V = Ah$. Cylinder: $V = \pi r^2 h$.

$r = 5,\ h = 8$: $V = 200\pi \approx 628.3$

11 Volume: pyramids and cones

$V = \frac{1}{3}Ah$: a third of the matching prism.

Cone $r = 3,\ h = 4$: $\frac{1}{3} \times 9\pi \times 4 = 12\pi \approx 37.7$

Square pyramid, base 6, height 10: $\frac{1}{3} \times 36 \times 10 = 120$

12 Spheres

$SA = 4\pi r^2$ and $V = \frac{4}{3}\pi r^3$

$r = 6$: $SA = 144\pi \approx 452.4$

$V = \frac{4}{3}\pi(216) = 288\pi \approx 904.8$

YEAR 10 MATHS · CHAPTER 5 · MIND MAP

Quadratic expressions and equations

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1 Expanding expressions

Multiply every term in one bracket by every term in the other.

$(x+3)(x-7) = x^2 - 4x - 21$

$(2x-5)^2 = 4x^2 - 20x + 25$

$(3x-1)(3x+1) = 9x^2 - 1$

2 Factorising expressions

Take out the HCF first, then look for the difference of two squares.

$12x^2 - 8x = 4x(3x-2)$

$25x^2 - 16 = (5x-4)(5x+4)$

$x^2 - 5 = (x-\sqrt{5})(x+\sqrt{5})$

3 Algebraic fractions: × and ÷

Factorise everything, cancel common factors, then multiply. To divide, flip the second fraction.

$\frac{x^2-9}{x+3} \times \frac{2}{x-3} = 2$

$\frac{x^2+2x}{4} \div \frac{x}{8} = 2(x+2)$

4 Monic trinomials

$x^2 + bx + c$: two numbers that multiply to c and add to b.

$x^2 - 2x - 15 = (x-5)(x+3)$

$x^2 + 9x + 20 = (x+4)(x+5)$

5 Non-monic trinomials

Find two numbers that multiply to ac and add to b, split the middle term, then group.

$3x^2 + 10x + 8$: $ac = 24$, use 6 and 4

$3x(x+2) + 4(x+2) = (3x+4)(x+2)$

6 Completing the square

Add and subtract $\left(\frac{b}{2}\right)^2$: $x^2 + bx = \left(x+\frac{b}{2}\right)^2 - \left(\frac{b}{2}\right)^2$

$x^2 + 6x + 4 = (x+3)^2 - 5$

$= (x+3-\sqrt{5})(x+3+\sqrt{5})$

7 Solving by factorising

Make one side zero, factorise, then use the Null Factor Law.

$2x^2 - 5x - 3 = 0 \Rightarrow (2x+1)(x-3) = 0$

$x = -\frac{1}{2}$ or $x = 3$

8 Applications of quadratics

Write the equation, solve it, then check which answers make sense.

A ball's height is $h = 20t - 5t^2$. When is it 15 m up?

$5t^2 - 20t + 15 = 0 \Rightarrow t^2 - 4t + 3 = 0$, so $t = 1$ or 3 s

Two answers: once going up, once coming down.

9 Solving by completing the square

Complete the square, then take the square root of both sides (±).

$x^2 - 4x - 1 = 0 \Rightarrow (x-2)^2 = 5$

$x = 2 \pm \sqrt{5}$

10 The quadratic formula

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$. The discriminant $\Delta = b^2 - 4ac$ gives the number of solutions.

$2x^2 + 3x - 1 = 0$: $\Delta = 9 + 8 = 17$, $x = \frac{-3 \pm \sqrt{17}}{4}$

$\Delta > 0$: two. $\Delta = 0$: one. $\Delta < 0$: none.

YEAR 10 MATHS · CHAPTER 6 · MIND MAP

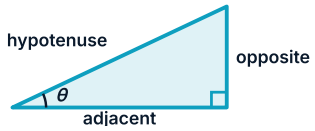
Trigonometry

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1 Trigonometric ratios

SOH CAH TOA.

Hypotenuse 12, angle 50°: opposite $= 12\sin 50^\circ \approx 9.19$



2 Finding unknown angles

Use the inverse buttons: $\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$.

$\cos\theta = \frac{5}{8} \Rightarrow \theta = \cos^{-1}(0.625) \approx 51.3^\circ$

3 Applications in 2D

Elevation and depression are measured from the horizontal.

From a 40 m cliff, the angle of depression to a boat is 25°

Distance $= \dfrac{40}{\tan 25^\circ} \approx 85.8$ m

4 Directions and bearings

True bearings: clockwise from north, three digits. Back bearing: add or subtract 180°.

N30°E = 030°T S40°W = 220°T

Back bearing of 060°T is **240°T**

5 Applications in 3D

Find the right triangle in the plane you need, often using Pythagoras first.

Box 3 by 4 by 12: base diagonal 5

Angle with the base: $\tan\theta = \frac{12}{5} \Rightarrow \theta \approx 67.4^\circ$

6 The sine rule

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$: use it with a side and its opposite angle.

$A = 40^\circ,\ B = 65^\circ,\ a = 10$: $b = \dfrac{10\sin 65^\circ}{\sin 40^\circ} \approx 14.1$

Watch out: finding an angle can give two answers (the ambiguous case).

7 The cosine rule

$c^2 = a^2 + b^2 - 2ab\cos C$: use it with two sides and the angle between, or three sides.

$a = 7,\ b = 9,\ C = 60^\circ$: $c^2 = 49 + 81 - 63 = 67$, so $c \approx 8.19$

8 Area of a triangle

$A = \frac{1}{2}ab\sin C$, with C between sides a and b.

$a = 7,\ b = 9,\ C = 60^\circ$: $A = \frac{1}{2} \times 63 \times \sin 60^\circ \approx 27.3$

9 The unit circle

A point on the unit circle is $(\cos\theta, \sin\theta)$.

$\sin(180^\circ - \theta) = \sin\theta$ $\cos(180^\circ - \theta) = -\cos\theta$

$\sin 150^\circ = \frac{1}{2}$ $\cos 120^\circ = -\frac{1}{2}$

10 Graphs of trig functions

$y = \sin x$ and $y = \cos x$: period 360°, between −1 and 1.

$\sin x$ starts at 0, $\cos x$ starts at 1

$y = \tan x$: period 180°, asymptotes at 90° and 270°

11 Exact values and trig equations

$\sin 30^\circ = \frac{1}{2},\ \cos 45^\circ = \frac{\sqrt{2}}{2},\ \tan 60^\circ = \sqrt{3}$

Solve $\sin x = \frac{1}{2}$ for $0^\circ \le x \le 360^\circ$: $x = 30^\circ$ or 150°

Watch out: check every quadrant where the ratio has the right sign.

YEAR 10 MATHS · CHAPTER 7 · MIND MAP

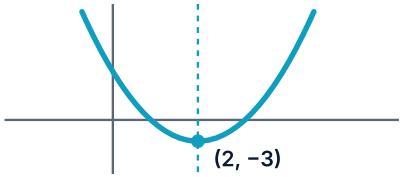
Parabolas and rates of change

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1 Exploring parabolas

$y = ax^2 + bx + c$: axis of symmetry $x = -\frac{b}{2a}$, y-intercept c.

$y = x^2 - 4x + 1$: axis $x = 2$, turning point $(2, -3)$



2 Sketching with transformations

$y = a(x - h)^2 + k$ has turning point (h, k).

$y = -2(x + 1)^2 + 3$: upside down, narrower, turning point $(-1, 3)$

3 Sketching using factorisation

Factorise for the x-intercepts. The turning point is halfway between them.

$y = x^2 - 2x - 8 = (x - 4)(x + 2)$

x-intercepts −2 and 4, axis $x = 1$, turning point $(1, -9)$

4 Sketching by completing the square

Complete the square to read off the turning point.

$y = x^2 + 6x + 5 = (x + 3)^2 - 4$

Turning point $(-3, -4)$

5 The quadratic formula and discriminant

$\Delta = b^2 - 4ac$ tells you how many x-intercepts.

$\Delta > 0$: two. $\Delta = 0$: touches once. $\Delta < 0$: none

$y = x^2 + 2x + 5$: $\Delta = 4 - 20 < 0$, no x-intercepts

6 Applications of parabolas

The maximum or minimum is at the turning point.

$h = -5t^2 + 20t + 1$: $t = 2$, max height 21 m

40 m of fence: $A = x(20 - x)$, biggest at $x = 10$: 100 m²

7 Lines and parabolas

Set the equations equal and solve. The discriminant tells you how many points they share.

$y = x^2$ and $y = x + 2$: $x^2 - x - 2 = 0$

$x = 2$ or -1: points $(2, 4)$ and $(-1, 1)$

8 Rates of change

Rate $= \frac{\text{change in } y}{\text{change in } x}$, in units per unit.

300 km in 4 hours: **75 km/h**

The gradient of a distance–time graph is the speed

9 Average and instantaneous rates

Average: gradient of the chord between two points. Instantaneous: gradient of the tangent at one point.

$y = x^2$ from $x = 1$ to 3: $\frac{9-1}{3-1} = 4$

At $x = 2$, using $x = 2.01$: $\frac{4.0401-4}{0.01} = 4.01 \approx 4$

10 Direct and inverse variation

Direct: $y = kx$. Inverse: $y = \frac{k}{x}$. Find k first.

$y \propto x$, $y = 12$ when $x = 3$: $k = 4$, so at $x = 5$, $y = 20$

$y \propto \frac{1}{x}$, $y = 6$ when $x = 2$: $k = 12$, so at $x = 4$, $y = 3$

YEAR 10 MATHS · CHAPTER 8 · MIND MAP

Probability and counting

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1 Review of probability

$P(A) = \dfrac{\text{favourable}}{\text{total}}$ and $P(A') = 1 - P(A)$.

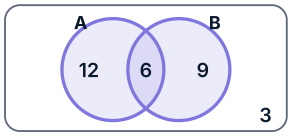
52 cards: $P(\text{heart}) = \frac{13}{52} = \frac{1}{4}$

$P(\text{not an ace}) = \frac{48}{52} = \frac{12}{13}$

2 Venn diagrams and two-way tables

$A \cap B$: both. $A \cup B$: either. A': not A.

30 people: $n(A \cup B) = 27$, $P(A') = \frac{12}{30} = \frac{2}{5}$



3 The addition rule

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$0.5 + 0.4 - 0.2 = 0.7$

Mutually exclusive: $P(A \cap B) = 0$

4 Conditional probability

$P(A \mid B) = \dfrac{P(A \cap B)}{P(B)}$: only look inside B.

From the Venn in card 2: $P(A \mid B) = \frac{6}{15} = \frac{2}{5}$

5 Arrays for two steps

List every outcome in a table.

Two dice: 36 outcomes. $P(\text{double}) = \frac{6}{36} = \frac{1}{6}$

$P(\text{sum} \geq 10) = \frac{6}{36} = \frac{1}{6}$

6 Tree diagrams

Multiply along the branches, add the branches you want.

2 red, 3 blue, take 2 without replacement:

$P(\text{one of each}) = \frac{2}{5} \cdot \frac{3}{4} + \frac{3}{5} \cdot \frac{2}{4} = \frac{3}{5}$

7 Independent events

Independent if $P(A \cap B) = P(A) \times P(B)$, so $P(A \mid B) = P(A)$.

Coin and die: $P(\text{H and } 6) = \frac{1}{2} \times \frac{1}{6} = \frac{1}{12}$

8 Counting and factorials

Multiply the number of choices at each step. $n! = n \times (n-1) \times \cdots \times 1$

3 shirts and 4 pairs of pants: 3 × 4 = 12 outfits

$5! = 120 \quad 0! = 1$

9 Arrangements

Order matters: ${}^nP_r = \dfrac{n!}{(n-r)!}$

Arrange 3 of 5 books: $5 \times 4 \times 3 = 60$

Arrange the letters of MATHS: $5! = 120$

10 Selections

Order doesn't matter: ${}^nC_r = \dfrac{n!}{r!(n-r)!}$

Choose 3 from 5: ${}^5C_3 = 10$

A pair from 6 people: ${}^6C_2 = 15$

Watch out: a team is a selection; a 1st, 2nd and 3rd place is an arrangement.

11 Counting in probability

$P = \dfrac{\text{favourable selections}}{\text{total selections}}$

Pick 3 from 5 boys and 4 girls: $P(\text{all girls}) = \dfrac{{}^4C_3}{{}^9C_3} = \dfrac{4}{84} = \dfrac{1}{21}$

YEAR 10 MATHS · CHAPTER 9 · MIND MAP

Statistics

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1 Collecting representative data

A sample should represent the whole population. Random samples avoid bias.

Census: everyone. Sample: some

Random, systematic or stratified sampling

2 Data displays

Histograms for numerical data, column graphs for categories.

Shape: symmetric, positively skewed (tail to the right), negatively skewed, bimodal

3 Two-way tables

Count by two categories, then use the row and column totals.

25 of 40 students play sport; 15 of those are in Year 10

$\frac{15}{25} = 60\%$ of the sport players are in Year 10

4 Summary statistics

Centre: mean and median. Spread: range and IQR.

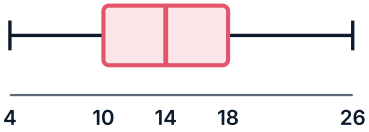
4, 6, 7, 9, 14: mean $\frac{40}{5} = 8$, median 7

An outlier pulls the mean, not the median.

5 Box plots and outliers

Outlier: below $Q_1 - 1.5 \times \text{IQR}$ or above $Q_3 + 1.5 \times \text{IQR}$.

$Q_1 = 10$, $Q_3 = 18$: IQR 8, upper fence $18 + 12 = 30$, so 34 is an outlier



6 Standard deviation

Standard deviation σ measures spread around the mean. Small σ: data close to the mean.

Bell-shaped data: about 68% within 1σ, 95% within 2σ

Mean 70, $\sigma = 5$: about 95% between 60 and 80

7 Cumulative frequency and percentiles

Cumulative frequency is a running total. Read percentiles off the graph (ogive).

Median = 50th percentile, Q_1 = 25th, Q_3 = 75th

40 values: the median is halfway between the 20th and 21st

8 Time-series data

Data collected over time, with time on the x-axis.

Look for the trend (up or down) and any seasonal pattern

9 Bivariate data and scatter plots

Two numerical variables: explanatory on the x-axis, response on the y-axis.

Describe the direction (positive or negative), strength and form (linear or not)

Watch out: correlation doesn't prove that one causes the other.

10 Line of best fit

Draw it by eye with about the same number of points on each side.

$y = 2.5x + 10$ at $x = 8$: $y = 30$

Predicting inside the data (interpolation) is more reliable than outside it

YEAR 10 MATHS · CHAPTER 10 · MIND MAP

Polynomials, functions and graphs

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1 Functions and notation

A function gives one y for each x. $f(a)$: substitute $x = a$.

$f(x) = 2x^2 - 3$: $f(2) = 5$, $f(-1) = -1$

Vertical line test: a function's graph is hit at most once by any vertical line.

2 Introducing polynomials

The degree is the highest power. The leading coefficient is the number in front of it.

$P(x) = 3x^4 - 2x + 7$: degree 4, leading coefficient 3, constant 7

3 Expanding and simplifying

Multiply every term by every term, then collect like terms.

$(x^2 + 2)(x - 3) = x^3 - 3x^2 + 2x - 6$

$(x + 1)^3 = x^3 + 3x^2 + 3x + 1$

4 Dividing polynomials

Long division: divide the leading terms, multiply, subtract, repeat.

$(x^3 + 2x^2 - 5x - 6) \div (x - 2) = x^2 + 4x + 3$, remainder 0

5 Remainder and factor theorems

Dividing $P(x)$ by $(x - a)$ leaves $P(a)$. If $P(a) = 0$, $(x - a)$ is a factor.

$P(x) = x^3 - 4x^2 + x + 6$: $P(2) = 0$, so $(x - 2)$ is a factor

$P(1) = 4$: dividing by $(x - 1)$ leaves 4

6 Solving polynomial equations

Factorise fully, then use the Null Factor Law.

$x^3 - 4x^2 + x + 6 = (x - 2)(x - 3)(x + 1) = 0$

$x = -1, 2$ or 3

7 Cubics $y = a(x - h)^3 + k$

The point of inflection is (h, k). A negative a flips it.

$y = (x - 1)^3 + 2$: point of inflection $(1, 2)$

8 Graphs of polynomials

Factors give the x-intercepts. A squared factor touches the axis; a cubed one flattens through it.

$y = (x + 1)(x - 2)(x - 3)$: x-intercepts −1, 2, 3, y-intercept 6

9 Graphs of circles

$(x - h)^2 + (y - k)^2 = r^2$: centre (h, k), radius r.

$(x - 2)^2 + (y + 1)^2 = 9$: centre $(2, -1)$, radius 3

10 Hyperbolas

$y = \frac{a}{x - h} + k$: asymptotes $x = h$ and $y = k$.

$y = \frac{2}{x - 1} + 3$: asymptotes $x = 1$ and $y = 3$

11 Further transformations

$f(x) + k$: up k. $f(x - h)$: right h. $-f(x)$: flip in the x-axis. $f(-x)$: flip in the y-axis.

$y = af(x)$: stretched from the x-axis by a factor of a